

Omar Tarek Farha

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EDUCATION

El Mahalla Official Languages School – Egypt (Expected Graduation: 2026) 2023 — Present
Relevant Coursework: Advanced Mathematics and Mechanics, Physics and Chemistry, Biology , Computer Science Fundamentals, French, and Advanced English.

PROFESSIONAL EXPERIENCE

Web Development Course (Self-Founded) Jun 2025 - Present
Founder & Instructor

- Founded and independently taught a Front-End development course for teenagers.
- Designed a curriculum covering HTML, CSS, and JavaScript.
- Taught and mentored **30+ students** in the first cohort, leading all sessions personally.
- Inspired students to pursue careers in coding and technology by applying hands-on learning.

Freelance Web Developer @ Xeno Company / WhiffWear store May 2025 - Present
Main Web Developer

- Designed and launched websites for clients, including an educational platform and e-commerce site.
- Applied responsive design and Full Stack development skills (Next Js, Supabase, Tailwind).
- Balanced freelance work with academic responsibilities, showcasing entrepreneurial initiative.

Online Learning Platform for Physics Teacher Aug 2025 – Present
Personal EdTech Project / Independent Developer

- Designed and developed an online learning platform for a private physics Teacher to manage lessons, exams, and student performance.
- Implemented a secure access code system — each student received a unique code to log in and access personalized materials and tests.
- Built features for uploading lessons, scheduling quizzes, and tracking progress, enhancing both the teacher's workflow and student engagement.

Ramy Gym Online Training Platform Sep 2025 – Present
Full-Stack Developer / Platform Co-Builder

- Built an online training platform for Ramy Gym that enables coaches to create and share personalized workout plans with clients.
- Developed dashboards where coaches can track client progress, schedules, and performance insights.
- Designed a responsive, user-friendly interface to enhance the training experience for both coaches and clients.

Celluplast Dashboard Project Nov 2024 – Jun 2025
Frontend Developer / Team Collaborator

- Developed a custom dashboard for the Celluplast team to manage project data, results, and performance metrics during competition.
- Streamlined team operations and presentation through an interactive interface that showcased their research and progress.
- The team won 2nd place and qualified to represent Egypt in the Malaysia ISEF competition.

Hack Club (STEM School) Nov 2024 - Oct 2025
Instructor

- Organized weekly coding sessions and tech meetups to encourage STEM engagement.
- Mentored peers in software development projects and collaborative coding tasks.
- Led innovation-focused events and competitions to introduce students to programming.

AWARDS AND COMPETITIONS

- **Afro-Asian Forum for Innovation and Technology in Malaysia (Oct 2025)**
1st Place (Gold Medal) in the Senior Software Leaders category. Competed against 25 teams from approximately 15 country and won first place.
- **Afro-Asian Forum for Innovation and Technology in Egypt (Apr 2025)**
1st Place (Gold Medal) in the Senior Software Leaderst category. Competed against 50-60 teams and won first place, qualifying to represent Egypt in the international competition (Oct 2025).
- **ScrapYard Hackathon'25 (Mar 2025)**
1st Place (Gold Medal), organized by the Ministry of Communications and the American Hack Club. **Chosen among the top winning teams nationwide, competing against 30–35 teams**, which qualified me for the Hackathon in the USA.

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- **Innovation and Technology Competition (ITC) – Egyptian Military Academy (Jun 2025)**
Competed among **30 national teams**, advancing to the **final third stage** after two successful online rounds, representing one of the top-performing teams in the competition.
 - **NASA Space Apps Challenge – Ismailia (Oct 2024)**
Participated in NASA’s global hackathon, developing an innovative space and Earth science solution as part of the Ismailia local competition.
 - **FIRST LEGO League (FLL) Robotics Competition (2018–2020)**
Participated for three consecutive years, earning multiple awards: **3rd place in Presentation (2018)**, **Judges’ Award (2019)**, and **2nd place in Robot Design (2020)**.

PUBLICATIONS

- Omar Farha, Mariam Islam Abdeldaim, Malak Hamama, et al.
Investigating Impact of Sound Frequencies and Light Exposure on Pepper Plant Growth, 17 October 2024, PREPRINT (Version 1) available at Research Square

TECHNICAL SKILLS

- **Frontend Development:** ReactJs, NextJs, Typescript, Tailwind CSS
- **Backend Databases:** NodeJs, MongoDB, Supabase
- **Programming Languages:** JavaScript, HTML, CSS
- **Design Prototyping:** Figma, Photosho
- **Other Tools:** Git, GitHub

PROJECTS

Ne3ma — Online Platform to Reduce Food & Material Waste Feb 2024 – Present
Founder & Full-Stack Developer

- Founded a social impact platform connecting companies, NGOs, and individuals to redistribute surplus goods — reducing food and material waste across Egypt.
- Designed and developed a full web system with an interactive map, AI-powered chatbot, verified user authentication, and real-time donation tracking to boost operational efficiency.
- Managed pilot testing with multiple users and organizations, achieving highly positive feedback and enabling seamless donation coordination through smart matching and automated logistics using **Next.js**, **Tailwind CSS**, **TypeScript**, **Supabase**, **Node.js**, **Google Maps API**, and **OpenAI API**.

AI Medical Voice Agent Jun – Aug 2025
Full-Stack Developer

- Developed a web-based healthcare voice assistant that enables patients to ask medical questions and receive instant AI-powered voice responses.
- Integrated **AssemblyAI** for real-time speech-to-text and text-to-speech processing, combined with **OpenAI API** for accurate and natural medical dialogue.
- Built and optimized the platform using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS**, ensuring high performance, accessibility, and responsive design.

Web Development Projects Nov 2024 – Present

- Designed and launched multiple web apps using **HTML**, **CSS**, **JavaScript**, **React**, **Next.js**, and **Tailwind CSS**.
- Built an **E-commerce platform** with offers tracking and smart purchase management features.
- Developed an **Educational platform** connecting teachers and students for online learning and progress tracking.

Agrosonic – Sound Waves for Sustainable Agriculture Aug - Oct 2024

- Innovated a method to enhance crop growth using controlled sound frequencies instead of chemical fertilizers.
- Designed a frequency-based growth model (300 Hz for early growth, 700 Hz for fruiting), tested on bell peppers.
- Conducted a 6-week-long experiment demonstrating improved nutrient uptake and yield without synthetic inputs.

References can be provided if needed